

```
import numpy as np
import pandas as pd
```

- Import Dataset

```
Sales = pd.read_csv(r"D:\intenship\Sales_transactions_2022_2025.csv")
Sales
```

	Transaction_ID	Order_ID	Customer_ID	Customer_Name	Customer_Age \
0	T02016694	00116694	C01482	Sophia Smith	28.0
1	T02016374	00108187	C01591	Hannah King	46.0
2	T02004099	00104099	C04140	Emma Wilson	42.0
3	T02010280	00110280	C03663	Daniel Clark	18.0
4	T02012685	00112685	C02749	Sophia Smith	37.0
...	...	...	...	...	...
18040	T02006253	00106253	C03487	Ella Adams	32.0
18041	T02005688	00102844	C00418	Jack Jones	32.0
18042	T02011672	00111672	C03311	Oscar Williams	37.0
18043	T02013094	00113094	C01590	Benjamin Moore	21.0
18044	T02011617	00111617	C01479	Lucas Allen	55.0
	Customer_Gender	Customer_Segment	Order_Date	Order_Time	\
0	Male	Consumer	2022-10-06	11:30:55	
1	Female	Consumer	2024-06-03	17:06:29	
2	Male	Consumer	2024-12-19	08:22:11	
3	Male	Corporate	2025-10-05	14:32:43	
4	Male	Returning Customer	2022-09-13	14:55:16	
...	...	...	...	...	...
18040	Male	New Customer	2022-01-11	10:44:18	
18041	Male	Returning Customer	2023-10-09	08:26:43	
18042	Male	Corporate	2025-04-02	17:45:09	
18043	Female	New Customer	2022-06-08	15:03:38	
18044	Male	Consumer	2025-06-30	19:01:38	
	Sales_Channel	...	Order_Status	Shipping_Method	Delivery_Days \
0	Online	...	Completed	Standard	4.0
1	Online	...	Completed	Standard	6.0
2	Online	...	Shipped	Standard	4.0

3	B2B Portal	...	Completed	Next Day	2.0
4	B2B Portal	...	Completed	Express	2.0
...	...	...	...	...	...
18040	Online	...	Completed	Express	2.0
18041	Retail Store	...	Processing	Standard	3.0
18042	Online	...	Completed	Standard	6.0
18043	Online	...	Completed	Standard	7.0
18044	Retail Store	...	Completed	Pickup	0.0

	Return_Flag	Return_Reason	Sales_Representative	Promotion_Code	\
0	No	NaN	James Anderson	SAVE10	
1	No	NaN	Ella Lewis	NaN	
2	No	NaN	Liam Davis	NaN	
3	No	NaN	Grace Harris	HOLIDAY20	
4	No	NaN	Sophia Brown	NaN	
...	...	...	...	...	...
18040	No	NaN	Emma Carter	MOM15	
18041	No	NaN	Emma Carter	SAVE10	
18042	No	NaN	Sophia Brown	SAVE10	
18043	No	NaN	Noah Wilson	BLACKFRIDAY25	
18044	No	NaN	Grace Harris	NaN	

	Customer_Rating	Inventory_Level	Order_Year
0	4.0	81.0	2022
1	5.0	123.0	2024
2	4.0	93.0	2024
3	NaN	59.0	2025
4	2.0	56.0	2022
...	...	...	...
18040	5.0	52.0	2022
18041	4.0	107.0	2023
18042	2.0	58.0	2025
18043	3.0	32.0	2022
18044	5.0	73.0	2025

[18045 rows x 36 columns]

- Basic summary

```
Sales.shape
```

```
(18045, 24)
```

```
Sales.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 18045 entries, 0 to 18044
```

```
Data columns (total 24 columns):
```

#	Column	Non-Null Count	Dtype
---	-----	-----	-----

```

0 Transaction_ID      18045 non-null object
1 Order_ID           18045 non-null object
2 Customer_ID        18045 non-null object
3 Customer_Name       18045 non-null object
4 Customer_Age        18045 non-null Int64
5 Customer_Gender     18045 non-null object
6 Customer_Segment    18045 non-null object
7 Order_Date          18045 non-null datetime64[ns]
8 Sales_Channel       18045 non-null object
9 Store_ID            18045 non-null object
10 Store_Name          18045 non-null object
11 Country            18045 non-null object
12 City               18045 non-null object
13 Product_ID         18045 non-null object
14 Product_Name       18045 non-null object
15 Product_Category   18045 non-null object
16 Quantity           18045 non-null int64
17 Unit_Price         18045 non-null float64
18 Discount_Percentage 18045 non-null float64
19 Sales_Amount       18045 non-null float64
20 Cost_Amount        18045 non-null float64
21 Profit             18045 non-null float64
22 Payment_Method     18045 non-null object
23 Order_Status       18045 non-null object
dtypes: Int64(1), datetime64[ns](1), float64(5), int64(1), object(16)
memory usage: 3.3+ MB

```

```
Sales.describe()
```

	Customer_Age	Order_Date	Quantity \
count	18045.0	18045	18045.000000
mean	38.917927	2024-02-11 06:20:43.690773248	2.127127
min	4.0	2022-01-01 00:00:00	-3.000000
25%	32.0	2023-02-13 00:00:00	1.000000
50%	39.0	2024-02-07 00:00:00	2.000000
75%	46.0	2025-02-13 00:00:00	3.000000
max	112.0	2025-12-31 00:00:00	11.000000
std	10.662362	NaN	1.786212

	Unit_Price	Discount_Percentage	Sales_Amount
Cost_Amount \			
count	18045.000000	18045.000000	18045.000000
mean	232.685633	6.256858	457.309027
min	21.390000	0.000000	16.680000
25%	53.380000	0.000000	78.530000
50%	99.870000	5.000000	168.810000

75%	285.580000	10.000000	472.450000	376.020000
max	1425.090000	110.000000	14303.070000	10791.000000
std	294.208022	6.881178	856.597181	705.866738

	Profit
count	18045.000000
mean	91.404131
min	-299.310000
25%	19.480000
50%	40.190000
75%	93.270000
max	3512.080000
std	167.827883

Cleaning Data

Check Datatype

Sales.dtypes

Transaction_ID	object
Order_ID	object
Customer_ID	object
Customer_Name	object
Customer_Age	float64
Customer_Gender	object
Customer_Segment	object
Order_Date	object
Order_Time	object
Sales_Channel	object
Store_ID	object
Store_Name	object
Country	object
Region	object
City	object
Product_ID	object
Product_Name	object
Product_Category	object
Product_Subcategory	object
Quantity	int64
Unit_Price	float64
Discount_Percentage	float64
Sales_Amount	float64
Cost_Amount	float64

```

Profit                float64
Payment_Method        object
Order_Status          object
Shipping_Method       object
Delivery_Days         float64
Return_Flag           object
Return_Reason         object
Sales_Representative  object
Promotion_Code        object
Customer_Rating       float64
Inventory_Level       float64
Order_Year            int64
dtype: object

```

```
Sales.columns
```

```

Index(['Transaction_ID', 'Order_ID', 'Customer_ID', 'Customer_Name',
      'Customer_Age', 'Customer_Gender', 'Customer_Segment',
      'Order_Date',
      'Order_Time', 'Sales_Channel', 'Store_ID', 'Store_Name',
      'Country',
      'Region', 'City', 'Product_ID', 'Product_Name',
      'Product_Category',
      'Product_Subcategory', 'Quantity', 'Unit_Price',
      'Discount_Percentage',
      'Sales_Amount', 'Cost_Amount', 'Profit', 'Payment_Method',
      'Order_Status', 'Shipping_Method', 'Delivery_Days',
      'Return_Flag',
      'Return_Reason', 'Sales_Representative', 'Promotion_Code',
      'Customer_Rating', 'Inventory_Level', 'Order_Year'],
      dtype='object')

```

- Drop Unwanted Columns

```
Sales.drop(columns=["Inventory_Level", "Promotion_Code"], inplace=True)
```

```
Sales.drop(columns=["Sales_Representative", "Return_Reason"],
inplace=True)
```

```
Sales.drop(columns=["Return_Flag", "Delivery_Days"], inplace=True)
```

```
Sales.drop(columns=["Order_Year", "Product_Subcategory"], inplace=True)
```

```
Sales.drop(columns=["Shipping_Method", "Region"], inplace=True)
```

```
Sales.drop(columns=["Order_Time", "Customer_Rating"], inplace=True)
```

```
Sales.columns
```

```

Index(['Transaction_ID', 'Order_ID', 'Customer_ID', 'Customer_Name',
      'Customer_Age', 'Customer_Gender', 'Customer_Segment',
      'Order_Date',

```

```
'Sales_Channel', 'Store_ID', 'Store_Name', 'Country', 'City',
'Product_ID', 'Product_Name', 'Product_Category', 'Quantity',
'Unit_Price', 'Discount_Percentage', 'Sales_Amount',
'Cost_Amount',
'Profit', 'Payment_Method', 'Order_Status'],
dtype='object')
```

- Change Data type

```
Sales["Order_Date"]=pd.to_datetime(Sales["Order_Date"],errors="coerce"
)
```

```
Sales["Customer_Age"]=Sales["Customer_Age"].astype("Int64")
```

```
Sales.dtypes
```

```
Transaction_ID      object
Order_ID            object
Customer_ID         object
Customer_Name       object
Customer_Age        Int64
Customer_Gender     object
Customer_Segment    object
Order_Date          datetime64[ns]
Sales_Channel       object
Store_ID            object
Store_Name          object
Country             object
City                object
Product_ID          object
Product_Name        object
Product_Category    object
Quantity            int64
Unit_Price          float64
Discount_Percentage float64
Sales_Amount        float64
Cost_Amount         float64
Profit              float64
Payment_Method      object
Order_Status        object
dtype: object
```

- Check duplicate and remove

```
Sales.duplicated().sum()
```

```
Sales.drop_duplicates()
```

	Transaction_ID	Order_ID	Customer_ID	Customer_Name
Customer_Age \				
0	T02016694	00116694	C01482	Sophia Smith

28					
1	T02016374	00108187	C01591	Hannah King	
46					
2	T02004099	00104099	C04140	Emma Wilson	
42					
3	T02010280	00110280	C03663	Daniel Clark	
18					
4	T02012685	00112685	C02749	Sophia Smith	
37					
...	...	...	...	...	.
..					
18040	T02006253	00106253	C03487	Ella Adams	
32					
18041	T02005688	00102844	C00418	Jack Jones	
32					
18042	T02011672	00111672	C03311	Oscar Williams	
37					
18043	T02013094	00113094	C01590	Benjamin Moore	
21					
18044	T02011617	00111617	C01479	Lucas Allen	
55					
	Customer_Gender	Customer_Segment	Order_Date	Sales_Channel	
Store_ID \					
0	Male	Consumer	2022-10-06	Online	
LA-01					
1	Female	Consumer	2024-06-03	Online	
SYD-01					
2	Male	Consumer	2024-12-19	Online	
PAR-01					
3	Male	Corporate	2025-10-05	B2B Portal	
NYC-01					
4	Male	Returning Customer	2022-09-13	B2B Portal	
CHI-01					
...	...	...	...	...	
...					
18040	Male	New Customer	2022-01-11	Online	
SYD-01					
18041	Male	Returning Customer	2023-10-09	Retail Store	
LON-01					
18042	Male	Corporate	2025-04-02	Online	
LON-01					
18043	Female	New Customer	2022-06-08	Online	
COL-01					
18044	Male	Consumer	2025-06-30	Retail Store	
TOR-01					
	Product_Name	Product_Category	Quantity	Unit_Price	\
0	...	Webcam HD	Electronics	2	85.29
...	...				

1	...	Office Keyboard	Electronics	6	58.85
2	...	Copy Paper Box	Office Supplies	2	43.92
3	...	SmartDock USB-C	Electronics	11	115.34
4	...	Desk Lamp	Furniture	4	69.71
...	...	...	...	...	...
18040	...	Portable Heater	Appliances	1	93.25
18041	...	AeroBook 14 Laptop	Electronics	2	901.65
18042	...	Security Camera	Electronics	1	159.93
18043	...	Notebook Pack	Office Supplies	2	27.20
18044	...	Backup Drive 2TB	Electronics	1	139.31

	Discount_Percentage	Sales_Amount	Cost_Amount	Profit	
Payment_Method \					
0	5.0	162.05	115.61	46.44	Credit
Card					
1	15.0	300.13	237.00	63.13	Debit
Card					
2	15.0	74.66	73.17	1.49	Credit
Card					
3	15.0	1078.46	897.91	180.55	
paypal					
4	15.0	237.03	164.95	72.08	
Apple Pay					
...	...	...	...	...	
...					
18040	0.0	93.25	66.55	26.70	Credit
Card					
18041	5.0	1713.14	1401.99	311.15	Credit
Card					
18042	10.0	143.93	116.02	27.91	
Apple Pay					
18043	0.0	54.41	32.71	21.70	Debit
Card					
18044	0.0	139.31	102.21	37.10	
PayPal					

	Order_Status
0	Completed
1	Completed
2	Shipped
3	Completed
4	Completed
...	...
18040	Completed
18041	Processing
18042	Completed
18043	Completed
18044	Completed

[18000 rows x 24 columns]

- Fill Null Values

```
Sales.isnull().sum()
```

```
Transaction_ID      0
Order_ID            0
Customer_ID         0
Customer_Name       108
Customer_Age        218
Customer_Gender     162
Customer_Segment    0
Order_Date          0
Sales_Channel       0
Store_ID            0
Store_Name          0
Country             0
City               0
Product_ID          0
Product_Name       144
Product_Category    0
Quantity            0
Unit_Price          0
Discount_Percentage 0
Sales_Amount        0
Cost_Amount         0
Profit              0
Payment_Method      147
Order_Status        0
dtype: int64
```

```
Sales["Customer_Age"]=Sales["Customer_Age"].fillna(Sales["Customer_Age"].median())
```

```
Sales["Payment_Method"]=Sales["Payment_Method"].fillna("Unknown")
```

```
Sales["Customer_Gender"]=Sales["Customer_Gender"].fillna(Sales["Customer_Gender"].mode())
```

```
Sales["Customer_Gender"]=Sales["Customer_Gender"].fillna("Unknown")
```

```
Sales["Product_Name"]=Sales["Product_Name"].fillna("Unknown Customer")
```

```
Sales["Customer_Name"]=Sales["Customer_Name"].fillna("Unknown Customer")
```

```
Sales.isnull().sum()
```

```
Transaction_ID      0
Order_ID            0
Customer_ID         0
Customer_Name       0
Customer_Age        0
Customer_Gender     0
```

Customer_Segment	0
Order_Date	0
Sales_Channel	0
Store_ID	0
Store_Name	0
Country	0
City	0
Product_ID	0
Product_Name	0
Product_Category	0
Quantity	0
Unit_Price	0
Discount_Percentage	0
Sales_Amount	0
Cost_Amount	0
Profit	0
Payment_Method	0
Order_Status	0
dtype:	int64